

1	a	Newton's Second law states that the <u>rate of change of momentum</u> of a body is <u>proportional/equal</u> to the <u>resultant force</u> acting on the body and takes place <u>in the direction of the resultant force</u> .	A1
	b(i)	$\rho = m/V$ $m = \rho V$ $= 1000 V$ $= 1000 (\pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2)$ $= 0.46 \text{ kg}$	C1 M1
	(ii)1.	$\Delta p = (\Delta m)v$ $= 0.46 \times 13$ $= 5.973 = 6.0 \text{ Ns}$	A1
	2.	$F = 5.973 / 0.20$ $= 29.865 = 30 \text{ N}$	C1 A1
	(iii)	Force on water by rocket/nozzle equal to force on rocket/nozzle by water but in the opposite direction	M1 A1
	(iv)	At 0.20 s, $m = 0.40 + 0.70 - 0.46 = 0.64 \text{ kg}$ $F = ma$ Force of water on rocket – weight = ma $30 - 0.64 g = 0.64 a$ $a = 37 \text{ ms}^{-2}$	C1 C1 A1
		Total:	11